# SET 1: MONTHLY SALES DATA – RSTUDIO

## RStudio Practical Record

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| **Course Code & Name** | DSA0601 – DATA HANDLING AND VISUALIZATION |

**DATASET**

| Month | Sales (in $) |
|---|---|
| January | 15000 |
| February | 18000 |
| March | 22000 |
| April | 20000 |
| May | 23000 |

## QUESTION 1 – LINE CHART: MONTHLY SALES

**AIM**

To create a line chart in RStudio to visualize the monthly sales trend.

**ALGORITHM**

1. Create vectors for months and sales.

2. Use plot() to draw the line chart.

3. Customize the X-axis with month names.

4. Add labels, title and grid.

5. Display the graph in the RStudio Plots pane.

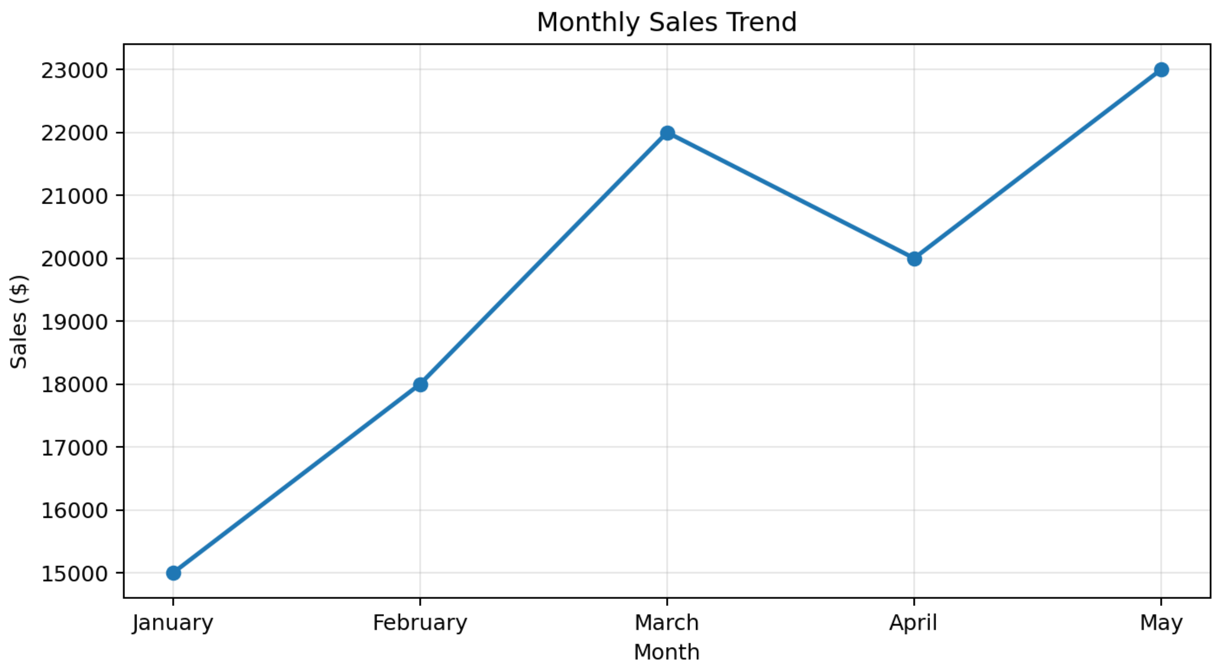
**PROCEDURE IN RSTUDIO**

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

**R PROGRAMMING CODE FOR RSTUDIO**

```
months <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

plot(1:5, sales,
     type = "o",
     pch = 16,
     xaxt = "n",
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales Trend")

axis(1, at = 1:5, labels = months)
grid()
```

**OUTPUT SCREENSHOT**




**RESULT**

Thus, the monthly sales trend was visualized successfully in RStudio using a line chart. Sales show an overall increasing trend with a small decline in April.

# QUESTION 2 – BAR CHART: TOP-SELLING PRODUCTS

### AIM

To generate a bar chart in RStudio showing the top-selling products.

### ALGORITHM

1. Create vectors for product names and units sold.

2. Use barplot() to create the chart.

3. Add axis labels and title.

4. Display the bar chart in RStudio.

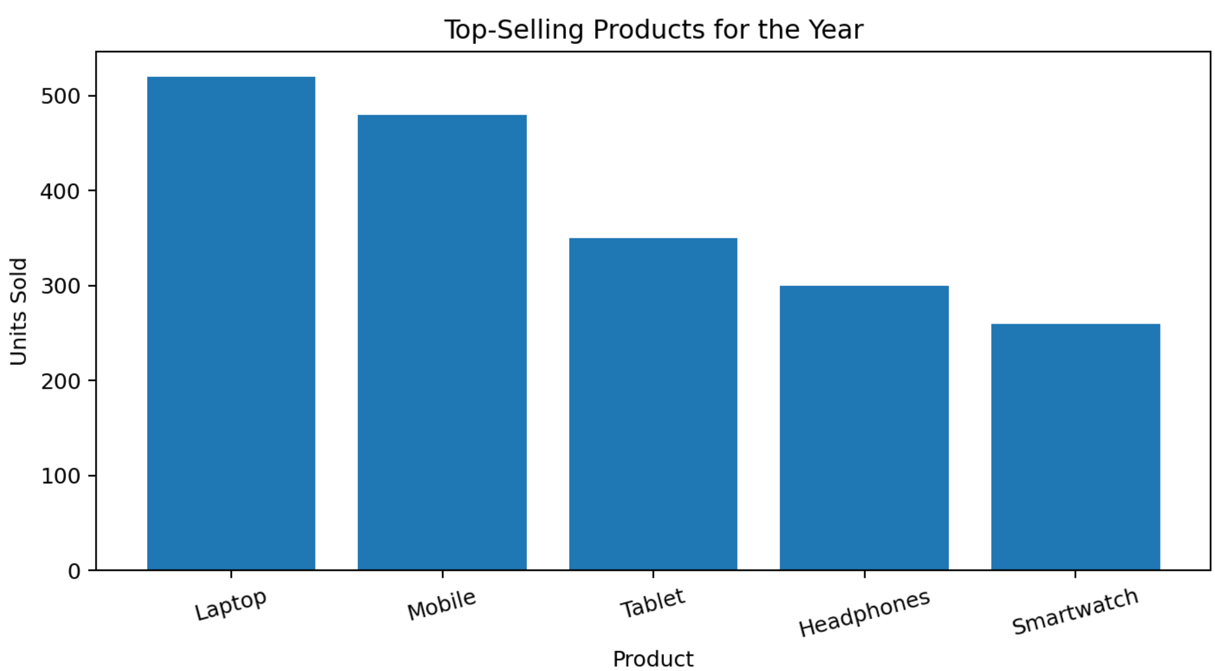
### PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

### R PROGRAMMING CODE FOR RSTUDIO

```
products <- c("Laptop", "Mobile", "Tablet",
              "Headphones", "Smartwatch")
units_sold <- c(520, 480, 350, 300, 260)

barplot(units_sold,
        names.arg = products,
        xlab = "Product",
        ylab = "Units Sold",
        main = "Top-Selling Products for the Year",
        las = 2)
```

### OUTPUT SCREENSHOT



**RESULT**

Thus, the top-selling products were displayed successfully using a bar chart in RStudio. Laptop has the highest sales in the sample data.

# QUESTION 3 – SCATTER PLOT: ADVERTISING BUDGET VS SALES

## AIM

To develop a scatter plot in RStudio to study the relationship between advertising budget and monthly sales.

## ALGORITHM

1. Create vectors for advertising budget and monthly sales.

2. Use plot() to create a scatter plot.

3. Add labels, title and grid.

4. Calculate correlation using cor().

5. Interpret the relationship.

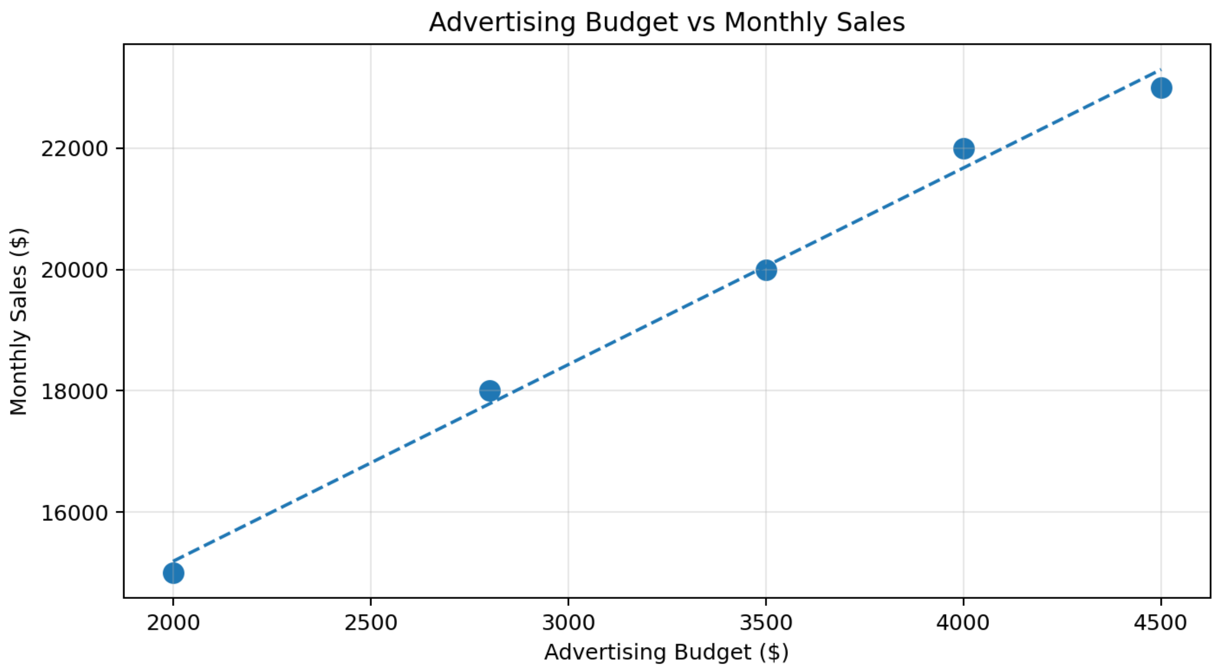
## PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

## R PROGRAMMING CODE FOR RSTUDIO

```
advertising_budget <- c(2000, 2800, 4000, 3500, 4500)
monthly_sales <- c(15000, 18000, 22000, 20000, 23000)

plot(advertising_budget, monthly_sales,
     pch = 16,
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     main = "Advertising Budget vs Monthly Sales")

grid()

correlation <- cor(advertising_budget, monthly_sales)
print(correlation)
```

## OUTPUT SCREENSHOT




**RESULT**

Thus, the scatter plot was created successfully in RStudio. The upward pattern shows a positive relationship between advertising budget and monthly sales.

# QUESTION 4 – INTERACTIVE SALES DASHBOARD IN RSTUDIO

### AIM

To build an interactive dashboard in RStudio by combining the monthly sales line chart and top-selling products bar chart.

### ALGORITHM

1. Install shiny and plotly packages.

2. Load the required libraries.

3. Create the sales and product data frames.

4. Design the Shiny user interface.

5. Create interactive Plotly charts in the server.

6. Run the Shiny application in RStudio.

### PROCEDURE IN RSTUDIO

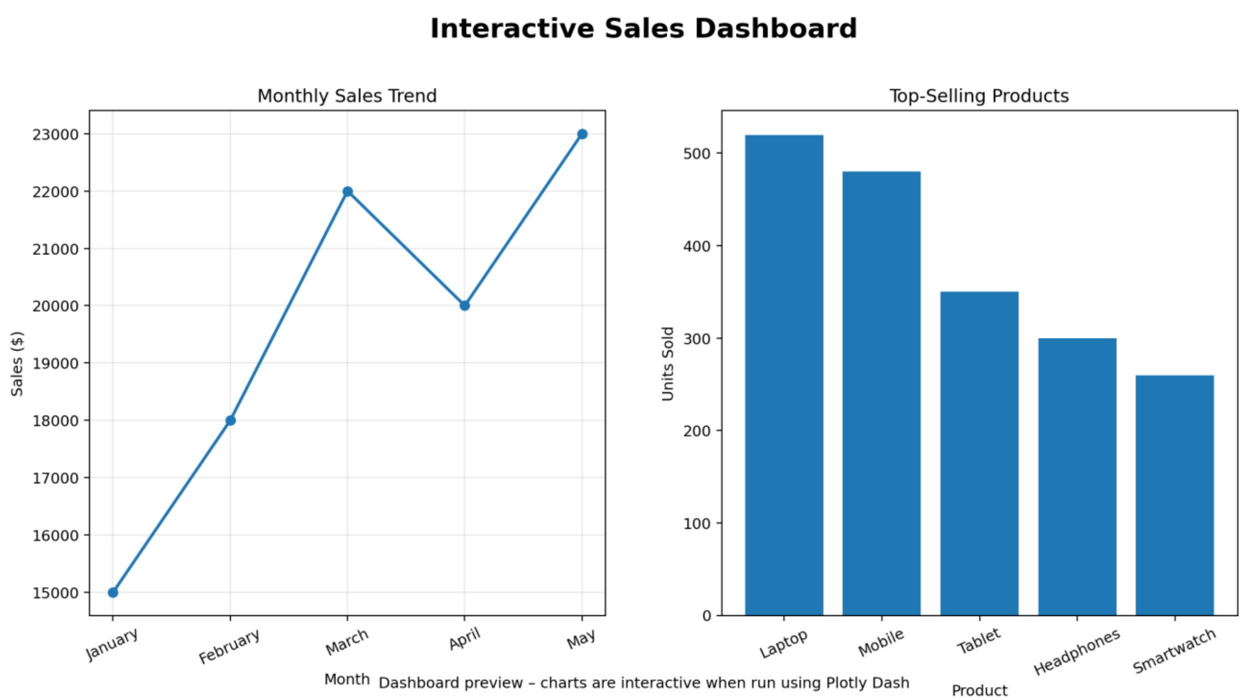
Open RStudio and create a new R Script. Run install.packages(c("shiny", "plotly")) once in the Console. Copy the code below into the script and save it as app.R. Click Run App at the top of the RStudio editor. The interactive dashboard opens in the RStudio Viewer or a web browser.

### R PROGRAMMING CODE FOR RSTUDIO

```
# Run this only once:
# install.packages(c("shiny", "plotly"))

library(shiny)
library(plotly)

sales_df <- data.frame(
  Month = factor(c("January", "February", "March",
                   "April", "May"),
                 levels = c("January", "February", "March",
                            "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

product_df <- data.frame(
  Product = c("Laptop", "Mobile", "Tablet",
              "Headphones", "Smartwatch"),
  Units_Sold = c(520, 480, 350, 300, 260)
)

ui <- fluidPage(
  titlePanel("Interactive Sales Dashboard"),
  plotlyOutput("lineChart"),
  plotlyOutput("barChart")
)

server <- function(input, output, session) {

  output$lineChart <- renderPlotly({
    plot_ly(sales_df,
            x = ~Month,
            y = ~Sales,
```

```
            type = "scatter",
            mode = "lines+markers") %>%
      layout(title = "Monthly Sales Trend")
  })

  output$barChart <- renderPlotly({
    plot_ly(product_df,
            x = ~Product,
            y = ~Units_Sold,
            type = "bar") %>%
      layout(title = "Top-Selling Products")
  })
}

shinyApp(ui = ui, server = server)
```

**OUTPUT SCREENSHOT**




**RESULT**

Thus, an interactive sales dashboard was developed successfully in RStudio using Shiny and Plotly.